

EXAM 2

STEP 1: SCHEDULE YOUR SESSION

Hyperlink: [Ordinary Differential Equation Exam](#)

STEP 2: WORK ON YOUR EXAM

THE EXAM HAS TWO SETS OF PROBLEMS, X and Y
CHOOSE ONLY ONE SET AND FOLLOW THE PROMPTS

STEP 3: SUBMIT AND SHOW UP

MAKE SURE THAT YOU FOLLOWED THE GUIDELINES

Guidelines¹

Written Part

- Your work must be submitted before your scheduled session, or handed in at the start of the session.
- Your scheduled session will not proceed if your written work has not been submitted or is incomplete.

Oral Part

- You must sign-up for a 15-minute time slot on exam days at least 24 hours in advance.
- If you had discussed an AES accommodation, you may schedule two consecutive sessions.
- You must show up to your scheduled oral part of the exam in-person and on time.
- A no-show or being late results in an automatic “M” (missed exam).

Submission

- Submit your work in Moodle before your scheduled session.
- If necessary, you may hand in a physical copy at the start of the session.

Collaboration and Academic Integrity Policy

- You may collaborate with your peers, but the work you submit must reflect your own.
- You may use all resources available to you, that includes your notes, books, and online resources.
- Be prepared to defend all aspects of your work.

Extra Credit Opportunity

- Optional problems will count as an extra credit added to your homework total.

¹Please review the course Syllabus for the complete Exam guidelines and assessment process.

X. Excessive Vibrations on a Robotic Vehicle

Description:

An autonomous delivery robot navigates uneven terrain and experiences vibrations due to external forces from the ground. The robot's suspension system on its wheels is modeled as a spring-mass-damper system. The vertical motion y at time t of the wheels is modeled by a linear 2nd-order ODE:

$$y'' + 2cy' + ky = f(t),$$

where c is the damping coefficient and k is the spring constant. The external forcing term representing road-induced vibrations is

$$f(t) = \sin(\beta t)$$

where β is the forcing frequency.

Written Part:

1. Show –by way of an *Ansatz*– that the characteristic equation is $r^2 + 2cr + k = 0$.
2. Show that the roots of the characteristic equation are

$$r_1 = -c + \sqrt{c^2 - k} \quad \text{and} \quad r_2 = -c - \sqrt{c^2 - k}.$$

3. Determine the homogeneous solution y_h for each of the following c and k parameter pairs.

a. $c = 0$ and $k = 2$

b. $c = 0$ and $k = 4$

c. $c = 1$ and $k = 2$

4. Use either the method of *Undetermined Coefficients* or *Variation of Parameters* to show that the particular solution y_p corresponding to each pair of c and k parameters matches one of the given solutions.

A. $\frac{\sin(\beta t)}{4 - \beta^2}$

B. $\frac{2 \sin(\beta t) - 2\beta \cos(\beta t) - \beta^2 \sin(\beta t)}{\beta^4 + 4}$

C. $\frac{\sin(\beta t)}{2 - \beta^2}$

5. For every pair of c and k values matched with each value of β , find both the general and specific solutions with initial conditions $y(0) = 1$ and $y'(0) = 0$.

i. $\beta = 2$

ii. $\beta = \sqrt{2}$

6. Use a graphing technology to draw clear and well represented graphs of the specific solutions determined in Problem (5).

Oral Part:

7. Explain the key steps of solving the homogeneous solution, and discuss why we need to determine this before solving the particular solution.
8. Explain the key steps of the chosen method for determining the particular solution, and provide a justification on why you can use such method.
9. Order the parameter sets –from lowest to highest– according to the maximum amplitude of the specific solutions. Then, interpret each solution in context of the robotic vehicle.

Optional:

10. Let $c = 2$, $k = 4$, and $f(t) = te^{-2t}$. Solve for the one independent homogeneous solution, find –by way of order reduction– another independent solution. Then, determine the particular solution.

Y. Resonance Effects in Radio Receivers

Description:

In a radio receiver, resonance is the principle that enables the circuit to selectively isolate and amplify a specific radio frequency signal from a wide range of signals. The circuit, with its current I at time t , is described by a linear 2nd-order ODE:

$$I'' + 2RI' + (1/C)I = F(t)$$

where R is the resistance coefficient and C is the capacitance constant. The external forcing term

$$F(t) = \cos(\alpha t)$$

is the rate of change of an electrical source with forcing frequency α .

Written Part:

1. Show –by way of an *Ansatz*– that the characteristic equation is $r^2 + 2Rr + (1/C) = 0$.
2. Show that the roots of the characteristic equation are

$$r_1 = -R + \sqrt{R^2 - (1/C)} \quad \text{and} \quad r_2 = -R - \sqrt{R^2 - (1/C)}.$$

3. Determine the homogeneous solution I_h for each of the following R and C parameter pairs.

$$\text{a. } R = 0 \text{ and } C = 2 \qquad \text{b. } R = 0 \text{ and } C = 1 \qquad \text{c. } R = 1 \text{ and } C = 1/2$$

4. Use either the method of *Undetermined Coefficients* or *Variation of Parameters* to show that the particular solution I_p corresponding to each pair of R and C parameters matches one of the given solutions.

$$\text{A. } \frac{2\alpha \sin(\alpha t) + 2\cos(\alpha t) - \alpha^2 \cos(\alpha t)}{\alpha^4 + 4} \qquad \text{B. } \frac{2\cos(\alpha t)}{1 - 2\alpha^2} \qquad \text{C. } \frac{\cos(\alpha t)}{1 - \alpha^2}$$

5. For every pair of R and kC values matched with each value of α , find both the general and specific solutions with initial conditions $I(0) = 1$ and $I'(0) = 0$.

$$\text{i. } \alpha = \frac{1}{2} \qquad \text{ii. } \alpha = \frac{1}{\sqrt{2}}$$

6. Use a graphing technology to draw clear and well represented graphs of the specific solutions determined in Problem (4).

Oral Part:

7. Explain the key steps of solving the homogeneous solution, and discuss why we need to determine this before solving the particular solution.
8. Explain the key steps of the chosen method for determining the particular solution, and provide a justification on why you can use such method.
9. Order the parameter sets –from lowest to highest– according to the maximum amplitude of the specific solutions. Then, interpret each solution in context of the radio receiver.

Optional:

10. Let $R = 1$, $C = 1$, and $f(t) = te^{-t}$. Solve for the one independent homogeneous solution, find –by way of order reduction– another independent solution. Then, determine the particular solution.